



CSE-476: Introduction to Natural Language Processing

Date: 01/22/2026

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation

Q1. In the BIO tagging scheme for NER, what does the “B-” prefix indicate?

- A. Between two tokens
- B. Beginning of an entity
- C. Inside an entity
- D. Outside any entity

Q2. Which statement best describes a (plain) Markov chain in sequence tagging?

- A. It uses only words (observations) and ignores labels
- B. It models transitions between labels but does not consider the words
- C. It models emissions $P(\text{word} \mid \text{label})$
- D. It is identical to an HMM

Q3. In a Hidden Markov Model (HMM) for NER, an emission probability is:

- A. $P(\text{label} \mid \text{previous label})$
- B. $P(\text{word} \mid \text{label})$
- C. $P(\text{label} \mid \text{word})$
- D. $P(\text{word} \mid \text{previous word})$

Q4. In an n-gram language model, a bigram LM predicts the next word using:

- A. No previous words
- B. The previous 1 word
- C. The previous 2 words
- D. The entire sentence history

Q5. Which statement about perplexity is correct?

- A. It's the raw probability of a sentence

- B. It's the inverse probability of a test text, normalized by length
- C. It's the fraction of correctly labeled tokens
- D. It's the number of unique words in the corpus